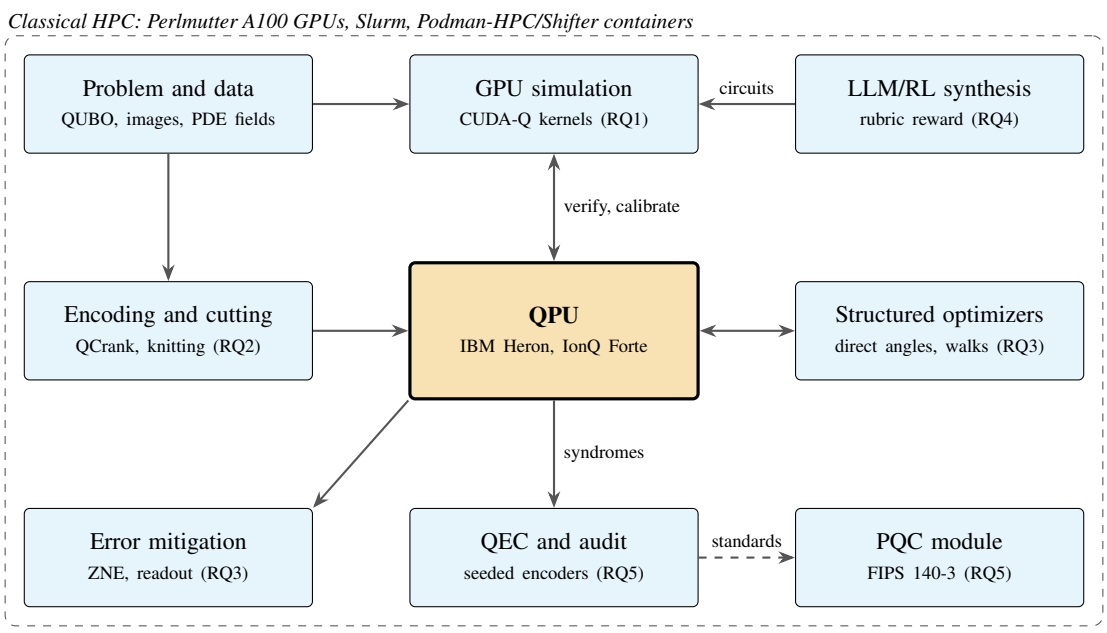

Classical HPC: Perlmutter A100 GPUs, Slurm, Podman-HPC/Shifter containers
Problem and data
QUBO, images, PDE fields
GPU simulation
CUDA-Q kernels (RQ1)
circuits
LLM/RL synthesis
rubric reward (RQ4)
verify, calibrate
Encoding and cutting
QCrank, knitting (RQ2)
QPU
IBM Heron, IonQ Forte
Structured optimizers
direct angles, walks (RQ3)
syndromes
Error mitigation
ZNE, readout (RQ3)
QEC and audit
seeded encoders (RQ5)
standards
PQC module
FIPS 140-3 (RQ5)